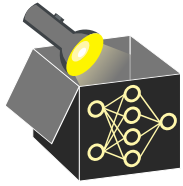
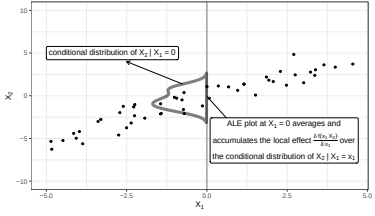
Interpretable Machine Learning



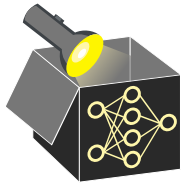
Feature Effects
ALE Walkthrough: Bike Sharing Example



Learning goals

- Step-by-step: partition, finite differences, average, accumulate, and center

FORMULA RECAP: WHAT ALE ESTIMATES



The formula for the **uncentered** first-order ALE at a specific point x is:

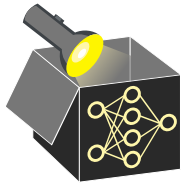
$$\hat{\tilde{f}}_{S,\text{ALE}}(x) = \underbrace{\sum_{k=1}^{k_S(x)}}_{\substack{\text{Step 4:} \\ \text{Accumulate}}} \underbrace{\frac{1}{n_S(k)} \sum_{i:\, x_S^{(i)} \in \text{int } k}}_{\substack{\text{Step 3:} \\ \text{Local Average}}} \underbrace{\left[\hat{f}(z_{k,S}, \mathbf{x}_{-S}^{(i)}) - \hat{f}(z_{k-1,S}, \mathbf{x}_{-S}^{(i)})\right]}_{\substack{\text{Step 1 \& 2:} \\ \text{Partition \& Finite Diff.}}}$$

- Dataset: bike rentals with features such as temperature, humidity, windspeed, season, weekday, etc.
- Model:

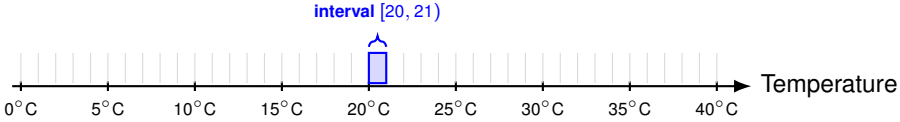
$$\hat{f}(\text{temp}, \text{humidity}, \text{windspeed}) = \text{predicted number of rented bikes}$$

- **Our Goal:** Explain the effect of **Temperature** (x_S) on the predicted number of rented bikes ($\hat{f}$), while keeping the interactions with Humidity and Windspeed ($\mathbf{x}_{-S}$) true to the data distribution.

STEP 1: PARTITION THE FEATURE RANGE

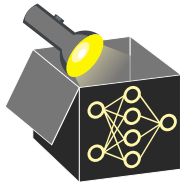


- Suppose temperature ranges from 0°C to 40°C in our training data.
- For this example, we split the feature space into $K = 40$ equidistant intervals of width 1°C ($[0, 1), [1, 2), \ldots, [39, 40)$).

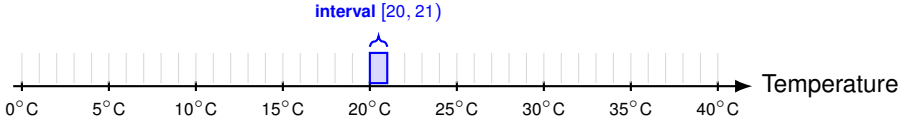


- Let's zoom in on the k-th interval: $[20°\mathbf{C}, 21°\mathbf{C})$.
- *Note:* Equidistant intervals preserve resolution, but quantile-based intervals may be used to ensure a balanced number of samples per interval.

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- Let's zoom in on the k-th interval: $[20°\mathbf{C}, 21°\mathbf{C})$.
- *Note:* Equidistant intervals preserve resolution, but quantile-based intervals may be used to ensure a balanced number of samples per interval.
- **Key intuition:** "Partitioning turns the feature range into small neighborhoods where effects can be estimated locally."

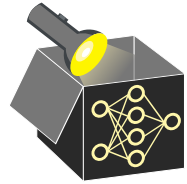
STEP 2: CALCULATE FINITE DIFFERENCES

- **1. Select:** Find a data point that falls into the interval (e.g., obs_A at 20.5°C).

Observation *A*

Temp = 20.5°C

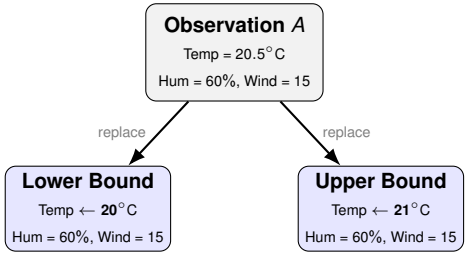
Hum = 60%, Wind = 15

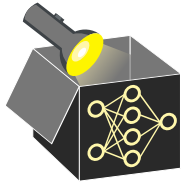


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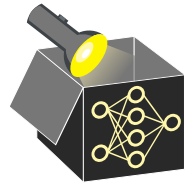
- **1. Select:** Find a data point that falls into the interval (e.g., obs_A at 20.5°C).
- **2. Substitute:** Replace its temperature with the lower (20) and upper (21) interval bounds.

  Crucial: Keep all other features exactly as observed!



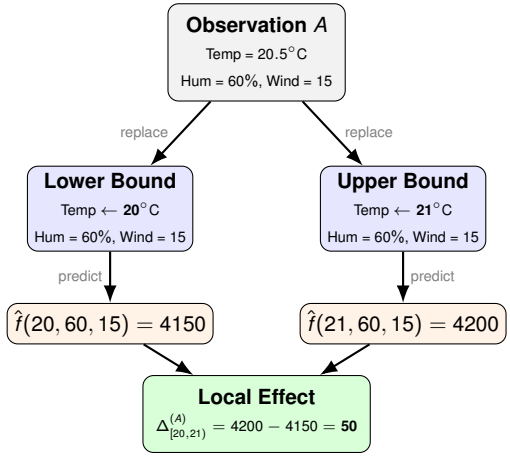


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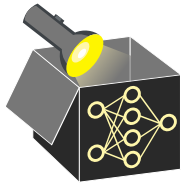


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- **3. Calculate:** Predict the outcomes for both new points and subtract to isolate the **local effect**.

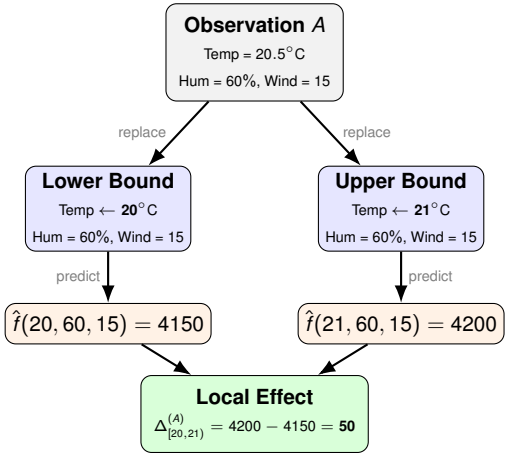


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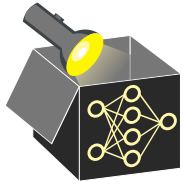
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- **Key intuition:** "What happens if we move temperature across this small interval?"

STEP 3: LOCAL AVERAGING



$$\bar{\Delta}_{[20,21)} = \frac{1}{n_{[20,21)}} \sum_{i:\, x^{(i)}_{\text{temp}} \in [20,21)} [\Delta^{(i)}_{[20,21)}]$$

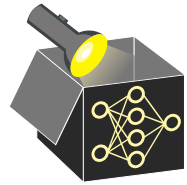
- We repeat Step 2 for **every** observation located inside the $[20, 21)$ interval.
- Because each observation has unique background features (humidity, windspeed), their individual local effects ($\Delta^{(i)}$) may vary.

Observation	Humidity	Windspeed	Finite Difference (Δ)
obs_A	60%	15 km/h	50 bikes
obs_B	64%	11 km/h	44 bikes
obs_C	58%	18 km/h	62 bikes
obs_D	70%	13 km/h	54 bikes

- By averaging these, we estimate the expected local effect: $\bar{\Delta}_{[20,21)} = \mathbf{52.5}$.

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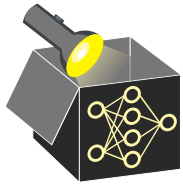


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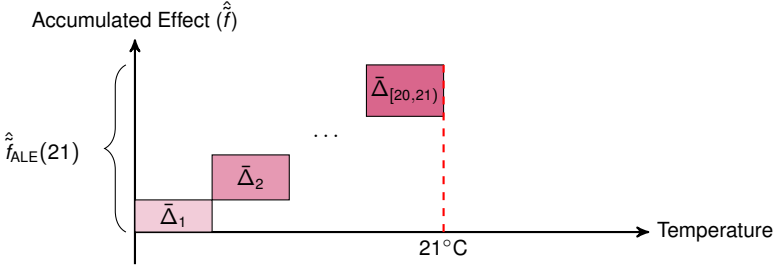
- By averaging these, we estimate the expected local effect: $\bar{\Delta}_{[20,21)} = \mathbf{52.5}$.
- **Key intuition:** "Average only over observations that actually live in this temperature region."

STEP 4: ACCUMULATE LOCAL EFFECTS

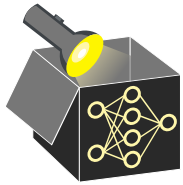


$$\hat{\tilde{f}}_{\text{temp,ALE}}(x) = \sum_{k=1}^{k(x)} \bar{\Delta}_k$$

To find the uncentered ALE effect at $x = 21°\text{C}$, we must **accumulate** (sum) all the average local effects from the minimum temperature up to 21°C.

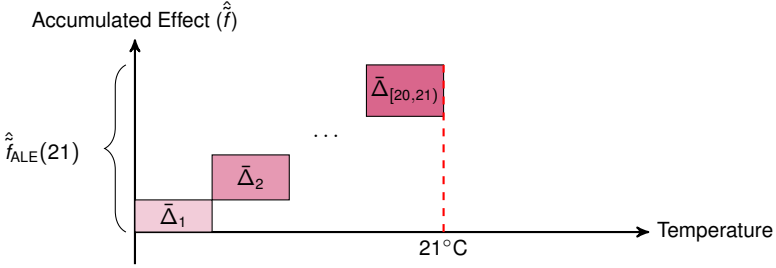


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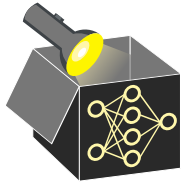
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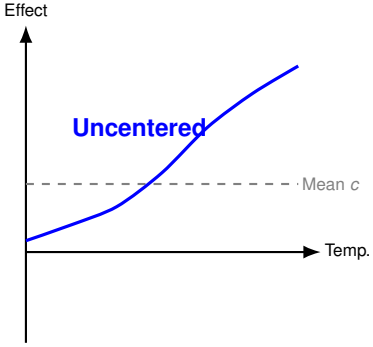
- **Key intuition:** "Add up these small local effects to recover the global temperature effect."

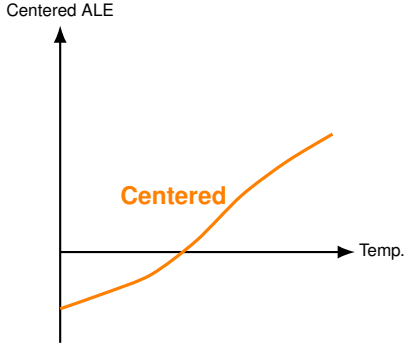
STEP 5: CENTERING



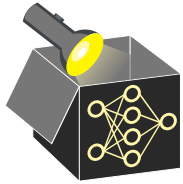
$$\hat{f}_{\text{temp,ALE}}(x) = \hat{\tilde{f}}_{\text{temp,ALE}}(x) - \underbrace{\frac{1}{n}\sum_{i=1}^{n}\hat{\tilde{f}}_{\text{temp,ALE}}(x_{\text{temp}}^{(i)})}_{\text{Mean shift } (c)}$$

Centering shifts the curve so the average effect across the dataset is 0.



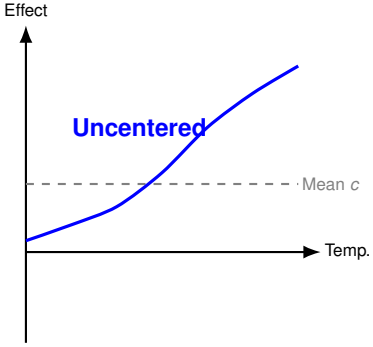


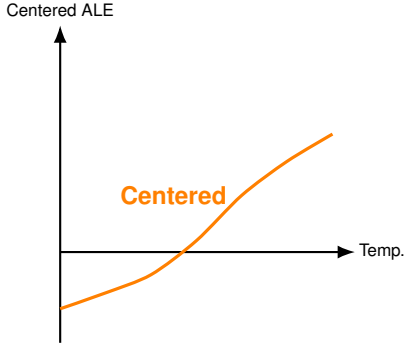
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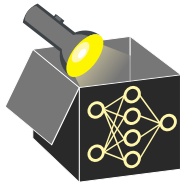
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- **Key intuition:** "Shift the curve so that it has mean zero."

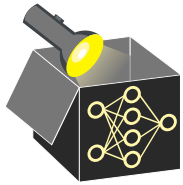
COMPLETE WALKTHROUGH IN ONE PICTURE



1. Partition temperature into intervals:

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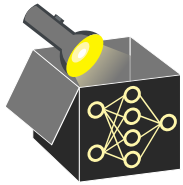
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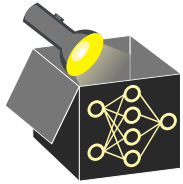
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3. Average these differences within the interval:

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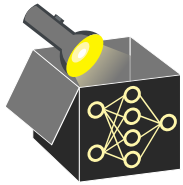
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5. Center:

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